# Chapter 3: Python Basics

## The Language of Financial Intelligence

On May 6, 2010, at 2:32 p.m. Eastern Time, the Dow Jones Industrial Average mysteriously plunged nearly 1,000 points in just minutes, wiping out almost $1 trillion in market value. This event, later known as the "Flash Crash," shocked financial markets. What caused this historic drop? Among other factors, investigators discovered that automated trading algorithms misinterpreting market signals played a significant role.

In those crucial moments, the market's fate was literally determined by code—specifically, the instructions that told computers when to buy and when to sell. Some algorithms worked as intended, while others spiraled out of control.

The Flash Crash taught the financial world a powerful lesson: in modern markets, understanding the language that speaks to computers isn't just a technical skill—it's the difference between prosperity and disaster.